

Binary Outcomes

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- Understand logistic regression and how to estimate it in R
- Interpret logit results using marginal effects and predicted probabilities
- Compare LPM and logit in practice

Roadmap

The Problem with Binary Outcomes

Logistic Regression

Interpreting Logit Results

Practice

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 - Did someone vote?
 - Did a war break out?
 - Did a bill pass?
 - Did a country democratize?
- Our outcome $Y \in \{0, 1\}$
- We want to model: $\Pr(Y = 1 \mid X)$

What happens if we just use OLS?

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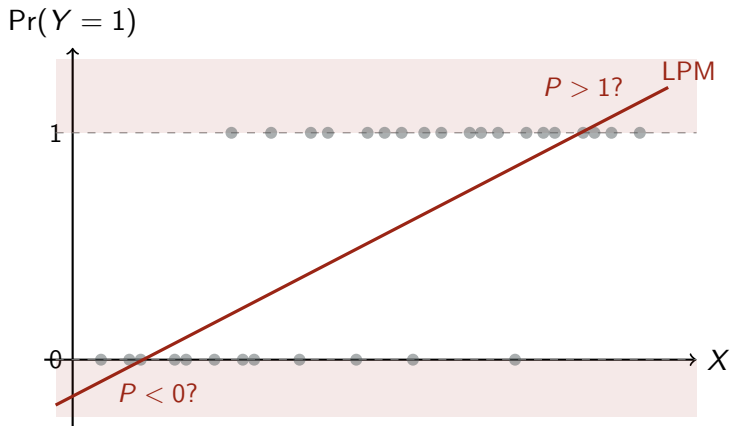
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- This is the **linear probability model** (LPM)

The LPM in pictures



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- Many applied researchers use the LPM in practice
- Especially common in economics and political science

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 - LPM forces it to be linear

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 - Rare events (many observations near 0)
 - You need predicted probabilities to be bounded
 - The relationship is clearly non-linear

You estimate an LPM predicting civil war onset. You find that 8% of predicted probabilities are negative.

Is this a problem? What would you do?

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The logistic function

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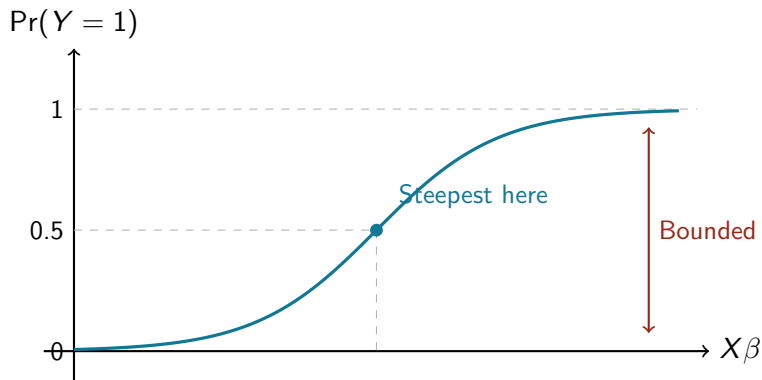
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- A natural model for probabilities

The sigmoid curve



The logit transformation

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- This is why we call it “logistic regression” or “logit”

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- The math is different, but the workflow is the same

Estimating logit in R

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- The syntax is identical to `lm()`, just swap to `glm()`
- Works with `broom::tidy()`, `modelsummary()`, etc.

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- The change in probability depends on where you start

Worked example: From log-odds to probability

$$\text{Logit model: } \log\left(\frac{P}{1-P}\right) = -2 + 0.5 \cdot \text{Education}$$

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- Going from 4 to 8 years: P goes from 0.50 to 0.88
- The same 4-unit change gives $\Delta P = 0.38$

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- We need better tools to interpret logit models

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Logistic Regression

Interpreting Logit Results

Practice

A logit model estimates $\hat{\beta}_1 = 0.8$ for education.

Your colleague says: “Education increases the probability of voting by 0.8.”

What’s wrong with this statement?

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- The most intuitive way to interpret logit models
- “What is the predicted probability of $Y = 1$ for a person with these characteristics?”
- In R:
 - `marginalEffects::predictions(model)`
 - Returns predicted probabilities for each observation
 - Or at specific values: `predictions(model, newdata = datagrid(...))`

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- In R: `marginaleffects::avg_slopes(model)`
- Interpretation (just like OLS):

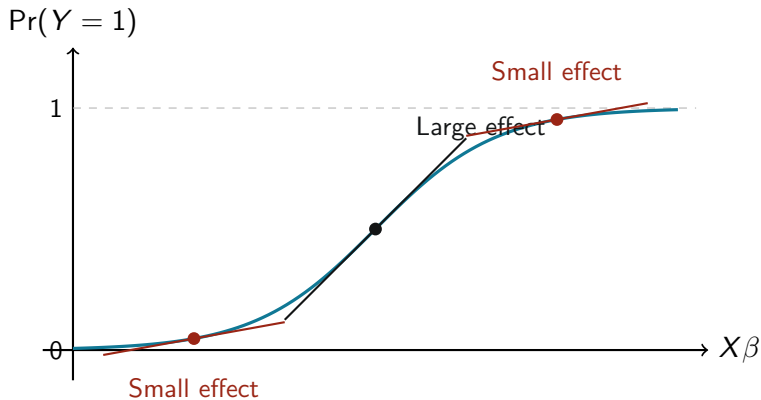
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- The marginal effect of X on $\Pr(Y = 1)$ varies across observations
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Why marginal effects vary



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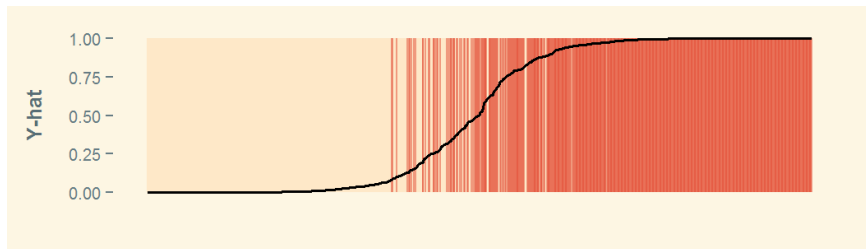
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- Reported automatically by `modelsummary()` and `performance::r2()`

Model fit for rare events: separation plot



- What are we really looking for in a binary model of rare events (e.g. civil wars, genocide, etc)?
- In R: `DIY` or `separationplot` package

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- Use whichever is conventional in your field

Roadmap

The Problem with Binary Outcomes

Logistic Regression

Interpreting Logit Results

Practice

Complete workflow in R

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avg_slopes(logit)                                # compare AMEs to LPM
plot_predictions(logit, condition = "income")
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 - You want to properly account for the binary nature of Y
- In practice: estimate both, report the one most appropriate
- Always report marginal effects, not just log-odds

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- The LPM is simple but has known limitations
- Logistic regression bounds probabilities between 0 and 1
- Log-odds are not intuitive — use marginal effects and predicted probabilities
- AMEs from logit are often similar to LPM coefficients
- Always estimate both and compare; always plot predicted probabilities

For next week

- Read Gelman et al., chapters 11–12
- Read Arel-Bundock, Greifer, and Heiss (2025), chapters 1–4
- Complete Assignment 3

- Next session: Model interpretation and diagnostics
 - Beyond coefficient tables
 - Visualizing model results
 - Residual diagnostics
 - Influence and outliers

Questions?